

# Four Routes to the Lepton Masses

An Inventory

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# **Four Routes to the Lepton Masses: An Inventory**

## Abstract

The FFGFT corpus now contains four independent approaches to the masses of the charged leptons: the T0 ladder from torus winding numbers (Docs. 006, 338, 352), the Koide formula with  $\theta = 2/9$  from a single  $A_5$  matrix element (Docs. 292, 367), the Galois identities from  $GF(3^n)$  orders (Doc. 338), and the  $\phi$ -skeleton of the icosahedral redistribution weights (Doc. 368). This document places the four routes side by side: input, output, accuracy against PDG, epistemic status. Finding: the routes deliver different *kinds* of statement (absolute masses, ratios, constraints) with accuracies from 1.6 % down to  $10^{-3}$  %. The Koide route is the only parameter-free one and the most precise; the T0 ladder the only one yielding absolute masses. A fifth route – the Koide scale  $M$  from the T0 ladder – is Route 1 summed (0.31 %) and shows that  $M$  is not an independent scale. The open edge is the same in all: the fractal-recursive correction from percent level to  $10^{-3}$  level is measured, not derived.

## Notation

### Epistemic markers:

- [B]** algebraically proved, exact arithmetic  
**[K]** numerically confirmed (approximation, accuracy stated)  
**[S]** structurally observed, derivation open  
**[E]** established external mathematics or physics

| Symbol            | Meaning                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------|
| $\xi$             | FFGFT fundamental parameter, $\xi = 4/30000 = 1/7500$                                                     |
| $v$               | electroweak scale, 246 GeV                                                                                |
| $r_i, p_i$        | winding quantum numbers of lepton $i$ (Doc. 338, Tab. 1)                                                  |
| $K_{\text{frak}}$ | fractal-recursive correction factor 74/75 (Doc. 338)                                                      |
| $\theta$          | Koide phase, $\theta = 2/9$                                                                               |
| $A$               | $\langle v_0   R_5 v_e \rangle$ , $ A  = \sqrt{2}/3$ (Doc. 367)                                           |
| $p_0, p_1, p_2$   | icosahedral redistribution weights (Doc. 293)                                                             |
| $\phi$            | golden ratio                                                                                              |
| PDG               | Particle Data Group 2024: $m_e = 0.51099895$ , $m_\mu = 105.6583755$ ,<br>$m_\tau = 1776.93 \pm 0.09$ MeV |

## 1 The Four Routes

### Route 1: T0 Ladder (Winding Numbers)

**Documents:** 006, 338, 352. **Input:**  $\xi$ ,  $v$ , quantum numbers  $(r_i, p_i)$ . **Output:** absolute masses.

$$m_i^{\text{bare}} = r_i \xi^{p_i} v, \quad \frac{m_j}{m_i} = \frac{r_j}{r_i} \xi^{p_j - p_i} \quad \textbf{[B]}$$

with  $(r_e, p_e) = (4/3, 3/2)$ ,  $(r_\mu, p_\mu) = (16/5, 1)$ ,  $(r_\tau, p_\tau) = (25/9, 2/3)$  from the torus winding numbers  $(n_\theta, n_\phi)$  (Doc. 338, Tab. 1).

| Quantity       | Formula                    | Value   | PDG     | Dev.    |
|----------------|----------------------------|---------|---------|---------|
| $m_\mu/m_e$    | $\frac{12}{5} \xi^{-1/2}$  | 207.846 | 206.768 | +0.52 % |
| $m_\tau/m_e$   | $\frac{25}{12} \xi^{-5/6}$ | 3531.6  | 3477.4  | +1.56 % |
| $m_e$ [MeV]    | $r_e \xi^{3/2} v$          | 0.5050  | 0.5110  | -1.18 % |
| $m_\mu$ [MeV]  | $r_\mu \xi v$              | 104.96  | 105.66  | -0.66 % |
| $m_\tau$ [MeV] | $r_\tau \xi^{2/3} v$       | 1783.4  | 1776.9  | +0.37 % |

**Status:** formula **[B]**, quantum numbers from  $\text{GF}(3^n)$  **[K]**. The residual deviation of 0.4–1.6 % is the fractal-recursive correction (Doc. 352): it is *measured* as  $+39.1 \xi$  ( $m_\mu/m_e$ ) and  $+117 \xi$  ( $m_\tau/m_e$ ; Doc. 352: 117.4 with older PDG value  $m_\tau = 1776.86$ ); its coefficients are not derived **[S]**.

### Route 2: Koide Formula with $\theta = 2/9$

**Documents:** 292, 367. **Input:** one matrix element  $A = \langle v_0 | R_5 v_e \rangle$ . **Output:** mass ratios; scale  $M$  external.

$$\sqrt{m_k} = M a_k, \quad a_k = 1 + 3|A| \cos(|A|^2 + \frac{2\pi k}{3}), \quad |A| = \frac{\sqrt{2}}{3} \quad \textbf{[B]}$$

The modulus of  $A$  supplies the coefficient  $\sqrt{2}$ , its modulus squared the phase  $\theta = 2/9$  (Doc. 367). No free parameter.

| Quantity                            | Formula                      | Value       | PDG      | Dev.      |
|-------------------------------------|------------------------------|-------------|----------|-----------|
| $m_\mu/m_e$                         | $(a_2/a_1)^2$                | 206.7703    | 206.7683 | 0.0010 %  |
| $m_\tau/m_e$                        | $(a_0/a_1)^2$                | 3477.47     | 3477.37  | 0.0031 %  |
| $m_\tau$ from $m_e, m_\mu, Q = 2/3$ |                              | 1776.97     | 1776.93  | 0.0022 %  |
| $Q_{\text{Koide}}$                  | $(\sum m)/(\sum \sqrt{m})^2$ | $2/3$ exact | 0.66666  | $10^{-5}$ |

**Status:**  $\theta = 2/9$  **[B]**.  $\cos(2/9)$  transcendental **[E]**, hence no algebraically exact ratios. For  $m_\tau/m_e$  the deviation ( $3.1 \cdot 10^{-5}$ ) lies *within* the PDG uncertainty of  $m_\tau$  ( $\pm 0.005\%$ ). For  $m_\mu/m_e$  this does not hold: at fixed amplitude  $\sqrt{2}$ ,  $(a_2/a_1)^2$  depends only on  $\theta$ , not on  $m_\tau$ ; against CODATA 206.7682827(46) the value 206.7703 lies rel.  $9.8 \cdot 10^{-6}$ , i.e.  $\approx 440\sigma$ , away. The exact pair  $(\sqrt{2}, 2/9)$  is excluded for  $m_\mu/m_e$  (as also stated in Doc. 292); compatibility would require an amplitude  $\sqrt{2} - 3 \cdot 10^{-7}$  at fixed  $\theta = 2/9$  (then  $m_\tau = 1776.97$  MeV,  $0.4\sigma$ ).

### Route 3: Galois Identities

**Document:** 338. **Input:** orders of  $\text{GF}(3^n)$ . **Output:** two constraints, not a complete determination.

$$\left(\frac{m_\mu}{m_e}\right)^2 = 43200 = 2^6 \cdot 3^3 \cdot 5^2, \quad m_e m_\mu = 54 \text{ MeV}^2 = |\text{GF}(3)^*| \cdot |\text{GF}(27)| \quad \textbf{[K]}$$

| Identity                       | Prediction | PDG     | Dev.    |
|--------------------------------|------------|---------|---------|
| $(m_\mu/m_e)^2 = 43200$        | 207.846    | 206.768 | 0.52 %  |
| $m_e m_\mu = 54 \text{ MeV}^2$ | 54         | 53.991  | 0.016 % |

**Status:** both **[K]**. The first identity is *the same number* as Route 1 ( $207.846 = \sqrt{43200}$ ): the T0 ladder reproduces the Galois identity exactly. For  $m_\mu/m_e$ , Routes 1 and 3 are not two routes but one route in two languages. The second identity is markedly more precise at 0.016 % and enters  $1/\alpha = 3700/27$  via  $\alpha = \xi E_0^2/K_{\text{frak}}$ .

### Route 4: $\phi$ -Skeleton of the $p$ -Values

**Document:** 368. **Input:**  $p_0, p_1, p_2$  from  $A_5$  (Doc. 293). **Output:** ratios algebraically, without cosine.

$$\frac{p_1}{p_2} = \phi^8, \quad \frac{p_0}{p_2} = 2\phi^4 \quad \textbf{[B]}$$

| Quantity     | Formula                            | Value   | PDG     | Dev.     |
|--------------|------------------------------------|---------|---------|----------|
| $m_\tau/m_e$ | $74 \phi^8$                        | 3476.42 | 3477.37 | 0.027 %  |
| $m_\tau/m_e$ | $74 \phi^8 (1 + \frac{27}{12}\xi)$ | 3477.47 | 3477.37 | 0.0029 % |
| $m_\mu/m_e$  | $30 \phi^4$                        | 205.62  | 206.77  | 0.55 %   |

**Status:**  $\phi$ -powers **[B]**, Galois factors 74, 30, 27/12 observed **[S]**. With the  $\xi$  correction,  $m_\tau/m_e$  reaches the precision of the Koide route.

## 2 Synopsis

| Route                 | Doc.        | Input                | Output      | $m_\mu/m_e$ | $m_\tau/m_e$ | Status                        |
|-----------------------|-------------|----------------------|-------------|-------------|--------------|-------------------------------|
| 1 T0 ladder           | 006/338/352 | $\xi, \nu, r_i, p_i$ | absolute    | 0.52 %      | 1.56 %       | <b>[K]</b> , corr. <b>[S]</b> |
| 2 Koide               | 292/367     | $A$                  | ratios      | 0.001 %     | 0.003 %      | $\theta$ <b>[B]</b>           |
| 3 Galois              | 338         | $\text{GF}(3^n)$     | constraints | 0.52 %      | —            | <b>[K]</b> , cause <b>[S]</b> |
| 4 $\phi$ -skeleton    | 368         | $p_j$                | ratios      | 0.55 %      | 0.027 %      | <b>[B]</b> + <b>[S]</b>       |
| 4b $\phi$ -sk.+ $\xi$ | 368         | $p_j, \xi$           | ratios      | —           | 0.003 %      | <b>[K]</b>                    |

### What each route delivers.

- **Route 1** is the only one yielding absolute masses – and the least accurate. Its value is the anchoring to  $\nu$  and  $\xi$ .
- **Route 2** is the only parameter-free one and the most precise. Its limit is the scale:  $M \approx 313.84$  MeV must come from outside. Only  $m_\tau/m_e$  lies within the PDG uncertainty of  $m_\tau$ ;  $m_\mu/m_e$  lies  $\approx 440\sigma$  from CODATA (see Route 2).
- **Route 3** yields no masses but two constraints. The more precise of the two ( $m_e m_\mu = 54$ ) is the anchor of the  $\alpha$  derivation.
- **Route 4** is the most algebraic: pure  $\phi$ -powers from the  $A_5$  geometry, no transcendental function. The price is the Galois factors, which are observed and not derived.

**Two accuracy classes.** Routes 1, 3 and 4a sit at *percent level* (0.5–1.6 %). Routes 2 and 4b sit at  $10^{-3}$ -*percent level*. For  $m_\tau/m_e$  both deviate by rel.  $\approx 3 \cdot 10^{-5}$ , within the  $m_\tau$  uncertainty ( $\pm 0.005$  %).  $m_\mu/m_e$  from Route 2 ( $(\sqrt{2}, 2/9)$ : 206.77032), by contrast, deviates by rel.  $9.8 \cdot 10^{-6}$ , i.e.  $442\sigma$  from CODATA 206.7682827(46), see Route 2; Route 4b does not give  $m_\mu/m_e$ . The difference between the classes is precisely the fractal-recursive correction:  $+117.4\xi$  in Route 1,  $+\frac{27}{12}\xi$  in Route 4, implicit in  $\cos(2/9)$  in Route 2.

## 3 The Common Open Edge

All four routes end at the same point. The percent class needs a  $\xi$ -proportional correction to become the  $10^{-3}$  class. This correction is known in three forms:

- as a measured coefficient 117.4 (Route 1, Doc. 352),
- as a Galois expression  $27/12 = |\text{GF}(27)|/(n_{\phi,\mu} n_{\theta,e})$  (Route 4, Doc. 368, 0.5 % agreement),
- as whatever the Koide formula contains beyond  $74\phi^8$  (Route 2).

A first-principles derivation is missing in all three forms **[S]**. The proof attempt via linearisation of  $\xi^{-5/6}$  (Doc. 368, Sect. 5) fails with 23 % discrepancy, because  $\xi^{-5/6}$  expands in  $\xi^{1/6}$ , not in  $\xi$ .

**Two readings of the correction.** Doc. 352 (§10) takes the T0 ladder as fundamental: the  $\varepsilon_i$  are distances between bare and PDG, decreasing with generation ( $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ ), and their origin is attributed to QED extraction and SI projection – as an attribution **[S]**, with the admission that the naive QED self-energy goes the wrong way. That is Reading A.

Doc. 367 permits a Reading B: the Koide formula with  $\theta = 2/9$  is fundamental, parameter-free from the icosahedron. The T0 ladder  $r_k \xi^{p_k}$  is then a *rational approximation* to the transcendental values  $a_k^2$  with  $a_k = 1 + \sqrt{2} \cos(2/9 + 2\pi k/3)$ . The residuals  $-39\xi, -77\xi, -116\xi$  emerge in Reading B without PDG and without QED – as the distance between cosine and

power law **[K]**. They differ because the three cosine values differ: a power law  $\xi^p$  has fixed slope  $\log \xi = -8.92$ , the Koide slopes  $2\Delta \log a/\Delta p$  are  $-10.66$  ( $e \rightarrow \mu$ ) and  $-8.47$  ( $\mu \rightarrow \tau$ ); the ratios  $r_j/r_i$  absorb the difference, the remainder is  $\varepsilon$ . No deeper reason is needed: the cosine is not a power law.

Both readings reproduce the same numbers. Reading B holds only with a restriction: the exact pair  $(\sqrt{2}, 2/9)$  is excluded for  $m_\mu/m_e$  ( $\approx 440\sigma$ , see Route 2). The readings differ in what they declare open. Reading A needs the R113 bridge (QED from  $T^4/\mathbb{Z}_3$ ). Reading B does not; its open question is instead: why are the rational approximation parameters exactly the  $\text{GF}(3^n)$  winding numbers, and why is the approximation so good (0.5%)? That is a question for number theory, not for QED. Reading B is the more economical one (Register R121).

**What distinguishes Route 2.** The Koide route does not need this correction separately: it is contained in  $\cos(2/9)$ . That is the reason for its precision – and the reason it is not algebraic. Routes 1, 3, 4 attempt to express the same result rationally or in  $\phi$ -powers; the transcendental remainder is the correction they then have to add back.

## 4 Route 5: The Koide Scale from the T0 Ladder

The Koide formula yields ratios; the scale  $M$  remains external. Can it come from Route 1?

**What  $M$  is.** With  $\sqrt{m_k} = \sqrt{M}a_k$  and  $\sum_k a_k^2 = 6$  one has exactly  $M = (m_e + m_\mu + m_\tau)/6 = 313.850$  MeV **[B]**. This is normalisation, not prediction. Equally exact:  $M = m_\tau/a_\tau^2$  with  $a_\tau^2 = 5.662$ .

**Finding.** Inserting the bare masses from Route 1:

$$M_{\text{bare}} = \frac{\nu}{6} \left( \frac{4}{3} \xi^{3/2} + \frac{16}{5} \xi + \frac{25}{9} \xi^{2/3} \right) = 314.82 \text{ MeV}, \quad \text{dev. } 0.31 \% \quad \mathbf{[K]}$$

Route 5 is Route 1 summed. The correction is  $+23.1 \xi$  and is arithmetically the weighted sum of the three individual corrections:  $-0.02 \xi$  ( $e$ )  $-2.78 \xi$  ( $\mu$ )  $+25.9 \xi$  ( $\tau$ ). It is  $\tau$ -dominated because  $M$  is 94%  $m_\tau/6$ .

**What this means.**  $M$  is not an independent scale. It follows from  $\xi$  and  $\nu$  with the accuracy of Route 1 – and with its open edge. A Galois-native single form  $M = c \xi^p \nu$  does not exist:  $M$  is a sum of three different  $\xi$ -powers ( $3/2, 1, 2/3$ ) that do not collapse into one (effective exponent 0.747, no rational coefficient for  $p \in \{2/3, 3/4, 5/6, 1\}$ ).

**Warning against number games.** Expressions such as  $M \approx 81\sqrt{15} \text{ MeV}$  (0.044%) or  $M \approx \frac{7}{40} \alpha \nu$  (0.1%) fit well but carry no mechanism. With the building blocks  $\{2, 3, 5, \phi, \sqrt{2}, \sqrt{3}, \sqrt{5}\}$  and exponents  $-4 \dots 4$ , twelve combinations hit  $M/E_0$  to within 0.1%;  $81\sqrt{15}$  is one of them, four others are better. The factor  $5/2$  in it appears in none of Docs. 292, 293, 338, 352, 367, 368. Such hits do not belong in the corpus.

## 5 What the Corpus Shows

Read correctly, the five routes tell a single story – and it is not about leptons but about what FFGFT predicts at all.

FFGFT computes in natural units. The fundamental parameter  $\xi = 4/30000$  is dimensionless, the winding numbers are integers, the Galois orders are integers, the icosahedral weights are algebraic numbers. None of this carries a unit. What follows from this structure is therefore *ratios*:  $m_\mu/m_e$ ,  $m_\tau/m_e$ ,  $Q = 2/3$ ,  $1/\alpha$ . And it is precisely these ratios that the corpus hits, wherever it computes parameter-free, to  $10^{-3}$  percent. Within experimental uncertainty lies  $m_\tau/m_e$ ;  $m_\mu/m_e$  from  $(\sqrt{2}, 2/9)$  deviates by rel.  $9.8 \cdot 10^{-6}$ , i.e.  $\approx 440\sigma$ , from CODATA (see Route 2). The Koide route needs one matrix element of the  $A_5$  geometry and nothing else. That is the theory's actual predictive power: the *shape* of the mass spectrum, not its size.

As soon as one demands a number in MeV, a scale must come from outside. FFGFT takes a single declared anchor for this – here  $\nu = 246$  GeV, elsewhere  $m_e$  or  $H_0$  – and this anchor serves conversion, not derivation. With the anchor one inherits the percent class: the T0 ladder yields absolute masses to 0.4–1.6 percent, and the Koide scale  $M$  is, as Route 5 shows, nothing but this ladder summed. The two accuracy classes of the synopsis are thus not two quality levels of the theory. They are the boundary between what FFGFT says of its own accord (ratios,  $10^{-3}$  %) and what it needs an external unit for (absolute values, percent).

The open edge lies exactly on this boundary. The fractal-recursive correction –  $117\xi$  here,  $27\xi/12$  there, hidden in  $\cos(2/9)$  for Koide – is the amount by which the rational and algebraic approximations deviate from the transcendental exact value. *Measuring* it is easy; *deriving* it from the winding numbers is what is still missing. Until then: FFGFT predicts the ratios of the lepton masses parameter-free at the  $10^{-3}$ -percent level (for  $m_\mu/m_e$  with the exact pair  $(\sqrt{2}, 2/9)$ , however, outside the measurement uncertainty), and their absolute values with one anchor at percent level. The one is theory; the other is theory plus a convention.